Physics 2 Problem Solving – Electric Field Part B – Total Charge

1) A positively charged straight wire with a uniform linear charge density of λ and length L.

   A) Determine the total charge on the wire if the linear charge density is a constant $\lambda = 2 \times 10^{-6}$ C/m and L = 0.5 m.

   Answer: 1 x 10^{-6} C



   B) Determine the total charge on the wire if the linear charge density changes with length given by $\lambda = (a)x^2$, assume "a" is a constant and $a = 4 \times 10^{-5}$ C/m^3. L = 0.5 m.

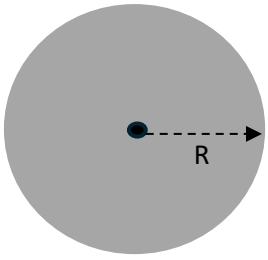
   Answer: 1.67 x 10^{-6} C

2) Using the semicircular thin wire with radius R = 0.2 m and a uniform linear charge density of $\lambda = 3 \times 10^{-6}$ C/m determine the total charge on the wire.



   Answer: 1.9 x 10^{-6} C

3) A circular disk with a radius R = 0.2 m and a non-uniform surface charge density that changes with r as $\sigma = (b)r$. Determine the total charge on the disk if the constant $b = 8 \times 10^{-5}$ C/m^3



   Answer: 1.34 x 10^{-6} C